
WordClock 5

Multi-Color LED Word Clock

User's Guide



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This guide describes the setup, configuration, and day-to-day operation of the WordClock 5 — a WiFi-connected, multi-color LED word clock built around the Angry Electronics software platform.

Revision/Update Information: Redesigned from V1.3

Clock Firmware: V1.3

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Table of Contents

Quick Start..... QS-1

1. Overview..... 1-1

2. Models..... 2-1

3. Display..... 3-1

4. Buttons..... 4-1

5. Power-On Functions..... 5-1

6. AP Configuration Mode..... 6-1

7. Live Settings Web Interface..... 7-1

8. Override Periods..... 8-1

9. Color Wheel Timer..... 9-1

10. NTP Time Sync..... 10-1

11. Announcements..... 11-1

12. Format Files..... 12-1

13. Over the Air Update..... 13-1

14. Troubleshooting..... 14-1

Quick Start

Follow these steps to set up the WordClock 5 for the first time. A full explanation of every option is in the numbered sections that follow.

1. Power on

Connect a 5V DC power supply to the clock. The barrel connector is 5.5/2.1 mm, center positive. Minimum current rating is 2 A for single-board clocks and 4 A for double- or triple-board clocks.

The clock is reverse-voltage protected, but it is NOT protected against higher voltages. A 12 V supply will most likely destroy every LED on every board. Verify the supply rating before plugging it in.

2. Enter AP configuration mode (automatic on first power-on)

A brand-new clock starts in AP configuration mode automatically. The display blinks "AP" to indicate it is waiting to be set up. (On a previously-configured clock that already has WiFi credentials, the clock proceeds directly to step 5 — to reconfigure WiFi, hold SW2 for 5 seconds while the time is displayed.)

3. Connect to the clock's WiFi

From a phone or computer, open the WiFi settings and connect to the network named WordClock-5. No password is required.

4. Configure in a browser

Open a web browser and go to 192.168.4.1. The setup page loads. At minimum:

- Select the home WiFi network from the SSID dropdown and enter its password.
- Confirm the UTC offset for the local region (US Eastern is 5 hours, Minus, by default).

Tap Save below the last field. The clock saves the settings and starts.

5. Wait for time to appear

On a successful boot, the display shows "WC5" briefly, then "WiFi" while connecting to the network, then "NTP" while fetching time from the internet, then the current time. The whole sequence usually completes within a few seconds.

6. Daily use

To do this	Do this
Change brightness temporarily	Click SW2 to cycle through ten brightness levels
Cycle through color effects	Click SW1 — effects cycle in the order shown in setup
Restore standard settings during an override	Click SW1 — original settings return for the Cancel Override Timer duration
See the clock's IP address	Double-click SW1 — IP and mDNS hostname scroll across the matrix
Adjust any setting live	Browse to the clock's IP address (or <a href="http://<ap-name>.local/">http://<ap-name>.local/) from any device on the same WiFi
Enter AP configuration mode	Hold SW2 for 5+ seconds while the time is displayed
Install new software over WiFi	Hold SW1 for 10 seconds while the time is displayed
Factory reset	Hold SW1 and SW2 while applying power; release when "WC5" appears
Format-file debug mode	Hold SW1 when "WC5" is displayed; release after the screen clears

7. Recommended next steps

- Browse the setup page on the live WiFi network and explore the available animations and LED effects.
- Configure Override Modes 1-3 to dim or change settings during sleeping hours.
- If using a multi-language or multi-board configuration, install an SD card with the appropriate format file.

1. Overview

The WordClock 5 displays the time in words, drawn from what appears at first glance to be a random matrix of letters. The matrix is eight rows deep and either sixteen, thirty-two, or forty-eight columns wide — built from one, two, or three sixteen-column boards connected together.

Display precision varies with the board count. The single-board version displays time to the nearest five minutes. The double-board version adds enough words to display time to the minute. The triple-board version additionally displays seconds.

Several languages are available. The clock animates the display in full color with a wide selection of effects and transitions, all selectable through a web setup page or with the two buttons on the clock.

WordClock 5 connects to internet NTP servers for time. Daylight Saving Time adjustments are automatic. An open-source format file is used for each version and each language, so users may adjust and customize the layout as they please.

Features

- Initial setup and configuration through a web browser on a PC or mobile device.
- UTC offsets including 30- and 45-minute time zones, for use anywhere in the world.
- Automatic Daylight Saving Time start and end with configurable rules.
- Connects to open or password-protected WiFi networks.
- Three independent override periods with their own animation, effect, and brightness settings.
- Optional SD-card announcements file for date- and weekday-based custom messages.
- User-editable format files on SD card for layout and language customization.

2. Models

Three model variants share this software platform. The model is determined by the number of display boards connected together.

Model	Boards	Matrix Size	Time Precision
WC5 Single	1	8 rows × 16 columns	Five minutes
WC5 Double	2	8 rows × 32 columns	One minute
WC5 Triple	3	8 rows × 48 columns	One second

The single-board English version has its format file built in. Other languages and the double- or triple-board configurations require an SD card containing the appropriate format file. See Section 12: Format Files.

3. Display

The clock displays time as illuminated words drawn from the letter matrix. Outside of normal time display, several short status messages appear on the screen:

Display	Meaning
WC5	Boot indicator. Shown briefly when the clock starts. The W should be red, C green, and 5 blue. If the colors appear in any other order, see Section 4: Buttons (SW2 power-on function) for the one-time correction procedure.
WiFi	Connecting to the configured WiFi network. If this message persists for more than a few seconds, the WiFi password is most likely incorrect.
NTP	Connecting to an NTP time server after the WiFi association has succeeded. Often shown so briefly it is missed.
AP	Access Point configuration mode. The clock is broadcasting its own WiFi network and waiting for setup.
SD Card	The SD card cannot be mounted. It is either missing, formatted incorrectly, or contains no readable format file. Only relevant for single-board clocks running a non-English language and for all double- or triple-board configurations.

On the triple-board version, seconds are part of the normal time display. On single- and double-board versions, the optional Color Wheel Timer animation provides a visual indication of seconds within the displayed minute (see Section 9: Color Wheel Timer).

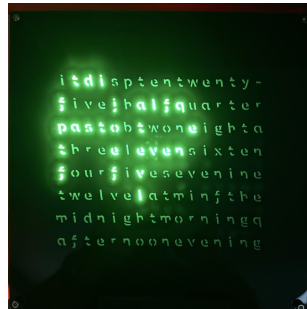
Status Display Examples



AP — Access Point configuration mode



WiFi — connecting to the configured network



NTP — fetching time from the time server

4. Buttons

The clock has two buttons: SW1 (left) and SW2 (right).

SW1 — During Normal Operation

Action	Function
Single click	Cycle through the configured color effects in the order shown in setup
Single click during an active override	Cancel the override and restore standard settings for the configured Cancel Override Timer duration
Double-click	Scroll the clock's IP address and mDNS hostname across the matrix (cyan IP, yellow hostname)
Hold 10 seconds	Start an over-the-air software update

SW2 — During Normal Operation

Action	Function
Single click	Cycle Nixie LED brightness through ten levels (temporary, not saved to memory)
Hold 5+ seconds	Restart in AP configuration mode. Existing settings are preserved so they can be edited.

Button Override

Pressing SW1 during an active override period temporarily restores the standard (non-override) settings. The restoration lasts for the duration set by Cancel Override Timer, then the active period's settings resume.

5. Power-On Functions

Special functions are available by holding buttons while applying power. These run before the clock initializes its normal button handling.

Buttons Held	Function
SW1 only	LED color cycle test. All LEDs cycle through red, green, and blue. Useful when checking soldering during assembly. Power-cycle the clock to exit.
SW2 only	Re-initialize display color order. Use if the boot "WC5" message does not show W=red, C=green, 5=blue. Release SW2 once "WC5" appears. Repeat if the colors are still wrong.
SW1 + SW2	Factory reset. Clears all saved settings and restarts in AP configuration mode. Release the buttons when "WC5" is displayed.
SW1 while "WC5" shown	Format-file debug mode. The clock advances time rapidly so format-file changes can be verified across many minutes and hours quickly. Release after the "WC5" message clears.

6. AP Configuration Mode

Entering AP Mode

- Hold SW2 for 5+ seconds while the time is displayed, OR
- Apply power with SW1 and SW2 both held (factory reset + AP mode), OR
- Power on a brand-new clock — it starts in AP mode automatically.

The display blinks "AP" to indicate the clock is in AP configuration mode.

Connecting

1. Connect a phone or computer to the WiFi network named WordClock-5.
2. Open a web browser and go to 192.168.4.1.
3. Adjust settings (described below) and press Save. The clock saves all settings and starts.

All hourly times use the 24-hour clock.

Network Settings

Setting	Description
Access Point Name	WiFi network name when in AP mode (default: WordClock).
Network SSID	Select the home WiFi network from the dropdown. SSIDs containing spaces may be displayed but can fail to connect; no more than 25 characters are recognized.
Password	WiFi password. Spaces within the password and trailing spaces are not allowed. Maximum length 25 characters.
NTP Server	Time server hostname (default: pool.us.ntp.org). See Section 10 for guidance on NTP server selection.

Display Settings

Setting	Options	Default
Animation	Selectable list of color animations	—

Animation Time	Frequency of the animation, or Off	—
LED Effect	Selectable color effect, or Off	—
LED Brightness	Brightness level, or Off	—

Override Periods 1, 2, and 3

Each of the three override periods has the same set of fields:

Setting	Description
Override Start Time	Hour at which the period begins (24-hour)
Override End Time	Hour at which the period ends (24-hour)
Override Days	All days, Weekdays only, or Weekends only
Animation	Animation to use during the period
LED Effect	LED effect to use during the period
LED Brightness	LED brightness to use during the period
Override Brightness	Display brightness during the period

When two or more override periods overlap in time, the period with the lowest number wins. See Section 8: Override Periods for use cases.

Time Zone

Setting	Options	Default
UTC Offset Hours	0–23	5
UTC Offset Minutes	0, 30, 45	0
UTC Offset Direction	Plus, Minus	Minus

Daylight Saving Time

Setting	Options	Default
DST Start Month	Jan–Dec	March
DST Start Week	1–5	2
DST Start Day	Sun–Sat	Sunday
DST Start Hour	0–23	2
DST Offset	Hours	1
DST Offset Direction	Plus, Minus	Plus
DST End Month	Jan–Dec	November
DST End Week	1–5	1
DST End Day	Sun–Sat	Sunday
DST End Hour	0–23	2

Default DST settings are configured for US Eastern Time.

Miscellaneous

Setting	Description
Announcement Color	Color of any announcement messages played from the SD card.
Cancel Override Timer	How long the standard settings are restored when SW1 is pressed during an active override.

7. Live Settings Web Interface

Once the clock is connected to WiFi, the live settings page is available from any device on the same network. All changes apply immediately as you adjust controls — no reboot — and a Save action persists changes across power cycles.

Access

Browse to `http://<clock-ip-address>/`. To find the IP address, double-click SW1 — the clock scrolls the IP (in cyan) followed by its mDNS hostname (in yellow) across the matrix.

Using a Name Instead of an IP Address (mDNS)

On home networks that support mDNS (also called Bonjour or zero-configuration networking), the clock can be reached by name instead of by IP. Browse to `http://wordclock.local/` (or whatever the configured Access Point Name resolves to) and the page should load. The hostname is derived from the AP Name set in AP configuration: lowercased, with spaces and any non-alphanumeric characters converted to hyphens. For example, an AP Name of "WordClock" becomes `wordclock.local`; "Kitchen Clock" becomes `kitchen-clock.local`. This makes it easy to give each clock in a multi-clock household a memorable, personal address.

mDNS does not work on every network. Some routers — particularly older models or routers with multicast filtering enabled — block the discovery messages mDNS depends on, in which case the `.local` address will not resolve. macOS, iOS, and Windows 10/11 generally work out of the box; some Android versions and Linux distributions need extra setup. If the `.local` address does not load, fall back to the IP address (double-click SW1 to display it).

Live Form Password

If a Live Form Password was set in AP configuration, the page prompts for it before any settings appear. Leave the password field blank in AP configuration to disable password protection.

Enabling and Disabling the Live Form

The live form can be turned off entirely from the Display page of AP configuration (Live Settings Form: Enabled / Disabled). When disabled, the clock does not serve any web

content while running — useful in shared environments where settings should not be editable after initial setup. Re-enabling requires another trip through AP configuration.

Layout

The live form is split across a Home page and two sub-pages:

- Home — page-navigation buttons, plus a "scroll a custom message" panel.
- Display — brightness, animation, animation interval, effect, and (if any override period has been configured) Override Mode enable/disable.
- Overrides — all three override periods plus the Cancel Override Timer.

Home Page — Scroll a Custom Message

Below the navigation buttons, three controls let you push an arbitrary text message onto the matrix as a one-shot scroll:

- Message Color — Red, Green, Blue, Yellow, White, or Custom Color
- Custom Color — color picker, used when Message Color is set to Custom Color
- Message — text field up to 63 characters, with a Send button that injects the message into the next available scroll cycle

Display Page

- Display Brightness — 0–9
- Animation — Falling Letters, Wandering Letters, The Matrix, Scroll, or Cycle
- Animation Interval — Off, 1 Minute, 5 Minutes, 10 Minutes, 15 Minutes, 30 Minutes, or 1 Hour
- Effect — Rainbow 1, Rainbow 2, Pulsing, Sweep, Twinkle, Red, Green, Blue, White, or Cycle
- Override Mode — Enabled / Disabled (only shown when at least one override period has been configured). Disabling cancels any active override and restores the standard settings until re-enabled.

Overrides Page

Each of the three override periods exposes the same fields as the AP setup form: Start Time, End Time, Brightness, Days, Animation, Animation Interval, Effect, and Custom Color. The Cancel Override Timer (15 Seconds, 1 Minute, or 5 Minutes) sits at the bottom of the page.

Saving Changes

Adjustments take effect on the matrix the moment a control is released — useful for trying out animations and effects without commitment. To make changes permanent, tap

Save at the bottom of the page; the clock writes the configuration to flash so it survives a power cycle. Reloading the page without saving leaves the running settings in place but does not persist them.

8. Override Periods

Three independent override periods schedule different display behavior at specific times of day. For example, the display can be dimmed at night or switched to a calmer effect during sleeping hours.

Each period has its own start time, end time, day mask (All / Weekdays / Weekends), animation, LED effect, LED brightness, and override brightness. All seven settings apply only while the period is active.

Priority: Period 1 > Period 2 > Period 3 when times overlap. Within a single period, all settings take effect together at the start time and reset together at the end time.

A short single click of SW1 during an active period restores the standard (non-override) settings for the duration of the Cancel Override Timer. After the timer expires, the active period's settings resume automatically.

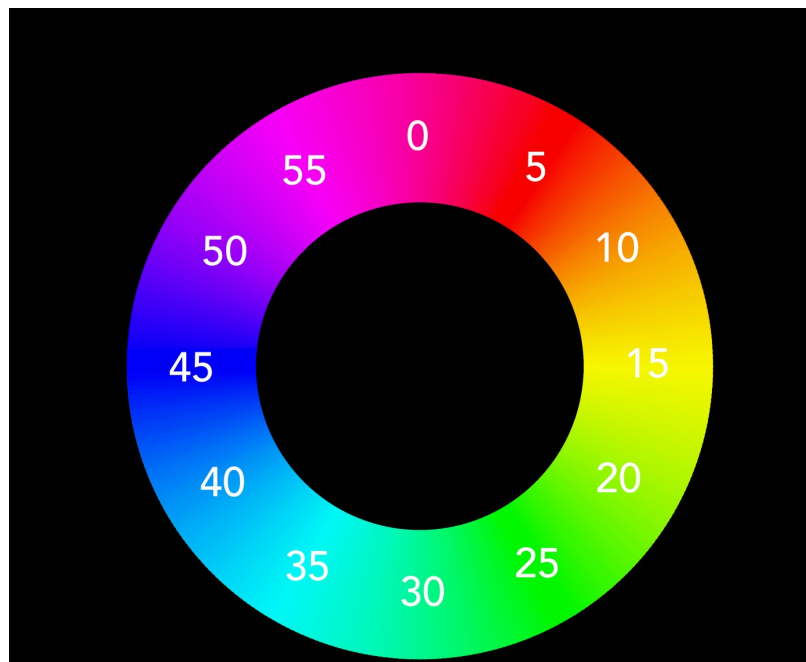
9. Color Wheel Timer

Color Wheel Timer is a special effect that cycles through a rainbow of colors at a fixed rate, providing a visual indication of time within the currently displayed minute or interval.

On single-board clocks (five-minute precision), the wheel completes one full rainbow cycle every five minutes. Once the user is familiar with the color order, the exact minute between the displayed five-minute marks can be read off the color of the wheel.

On double-board clocks (one-minute precision), the wheel completes one full cycle every minute, allowing seconds to be estimated.

On triple-board clocks (one-second precision), seconds are already shown directly on the matrix, so the Color Wheel Timer adds no additional information. It is still available as a decorative effect, and one full rainbow cycle still corresponds to one minute.



Color positions across one full rainbow cycle

10. NTP Time Sync

WordClock 5 obtains the current time from internet NTP (Network Time Protocol) servers. After WiFi associates on boot, the clock contacts the configured NTP server and uses the result to set its internal clock. Daylight Saving Time is then applied automatically from the configured DST rules.

Choosing an NTP Server

The default NTP server is `pool.us.ntp.org`. The NTP Pool Project terms of service ask end users not to make more frequent requests than the pool agreement allows:

"4. End-User agrees that he or she will not:

(a) Change default settings to make more frequent request of the Services, if using an ntp daemon ("NTP Protocol");

(b) Request time from the Services more than once every thirty (30) minutes (ideally at even longer intervals), if using SNTP." (<https://www.ntppool.org/tos.html>)

A single WordClock 5 polls infrequently and sits well inside this limit, but a household with many clocks all pointed at the same public pool can drift out of compliance. In that case, point each clock at a local time server instead.

Local NTP Servers

Most modern home routers run a built-in NTP server that any device on the home network can use as its time source. The exact address depends on the router; common values are `192.168.1.1` or `198.162.1.1`. Consult the router's documentation or admin page to confirm the address.

Other publicly available time servers — for example, `time.cloudflare.com` or `time.google.com` — are also free to use and have their own usage policies. A web search will identify many.

11. Announcements

The announcements file is an optional feature that schedules custom messages tied to specific calendar dates or recurring weekday occurrences. Announcements are read from the SD card and displayed at the 30-second mark of each minute on the day they apply.

File Location

The file must be saved at this exact path on the SD card:

`/Announce/Announce.txt`

It is a plain text file. Any standard editor (Notepad, TextEdit, etc.) can be used to create or modify it. The file is not included with the standard library distribution and must be created manually.

Limits

- Up to 75 announcements per file, one per line.
- Each announcement is at most 62 characters long.
- Entries can appear in any order — the file does not need to follow calendar sequence.

Date-Based Announcements

A date-based entry consists of two digits for the day, a separator (dash or forward slash), two digits for the month, then the announcement text. Example:

06-20Happy Birthday Melissa

Use 00 in place of the month to repeat the entry on the same day of every month.
Example:

00-12Pay Estimated Taxes

This shows the reminder on the 12th of every month.

Weekday-Occurrence Announcements

A weekday-occurrence entry triggers on a specific weekday and a specific occurrence within the month — for example, "the last Monday of May." The structure is: two-digit

month, separator (dash or slash), weekday index (Sunday=1, Monday=2, ..., Saturday=7), comma, occurrence number (1–5 for the first through fifth occurrence, or > for the last occurrence in the month), then the announcement text.

Examples

Entry	Triggers on
05-01,>Memorial Day	Last Monday of May, every year
03/02,2Second Tuesday	Second Tuesday of March, every year
10,0,3Third Sunday	Third Sunday of October, every year

12. Format Files

Format files define which LEDs light up to spell each word at each time. Only the single-board English version has its format file built into the firmware; every other configuration — non-English single-board, all double-board, all triple-board — requires an SD card holding a matching format file.

Note: support is not provided for changing or developing new format files. Users who modify their own format files do so at their own risk; the format-file debug mode (Section 5) helps verify changes by advancing time rapidly across many minutes and hours.

SD Card Layout

- Folder name: Language
- File extension: .fmt
- File name: at most eight characters before the extension (e.g. English5.fmt)
- Only one .fmt file may live in the Language folder at a time. Additional .fmt files can be stored elsewhere on the SD card for archival.

The SD-card format file overrides the built-in format file when present.

File Structure

A format file has six sections. The first five — Prefix, Minutes, Separator, Hours, and Suffix — define which LEDs make up each phrase. The sixth section, Format, controls when each phrase is displayed.

Syntax rules:

- Comments start with // and run to end of line.
- Each section header ends with a colon and a semicolon, e.g. Minutes;;
- Each definition or command line ends with a semicolon.

Phrase Definitions (sections 1–5)

Each definition has the form: tag, line, row, length;

For Minutes and Hours, the tag must match the corresponding numeric time. For example, in English5.fmt, "5,2,1,4;" defines the five-minute mark — line 2, row 1, four LEDs long — and lights the letters "f i v e."

For Prefix, Separator, and Suffix, the tag is any alphanumeric character (case-independent) used as a label by the Format section.

Format Section (section 6)

Each Format line has the form: section, tag, time-to-reference, >=, <=;

section names a phrase section (prefix, minutes, separator, hours, suffix). tag selects a specific phrase within that section, or is left blank to let the firmware pick the phrase that matches the current time. time-to-reference is one of: minutes, minute10, minute5, minute1, hours, hours10, hours1, time. The >= and <= columns limit the line to a specific range of the chosen reference.

Two special tag values exist:

- + — display the phrase as part of "to the next hour" wording. Example from English5.fmt: prefix,+,35,59;
- - — subtract one hour from the currently referenced time. Useful only on multi-board clocks where a second language section needs to roll back its hour reference.

A blank tag forces a search of the named section for the phrase whose tag matches the current reference time digits.

English1.fmt — Annotated Example

Comments in the file describe the phrase each definition spells. The first three sections (Prefix, Minutes, Separator) define the words "it / is" and the minute words and dashes; the Hours and Suffix sections define the hour words and the time-of-day suffixes ("o'clock", "in the morning", "in the afternoon", "in the evening", "midnight"). The Format section then chains them together: it sets the prefix, picks a minute phrase from the current minute, prints "minutes past", picks an hour phrase from the current hour, and chooses one of the four time-of-day suffixes based on the hour.

The complete file is short and worth reading start to finish — see English1.fmt in the project distribution.

13. Over the Air Update

The clock can update its software over WiFi — no USB cable, no programming tools required. To trigger an update, hold SW1 for 10 seconds while the time is displayed. The clock contacts the firmware server, compares versions, downloads the new image if needed, flashes it, and reboots.

The default firmware URL is <https://mitsf.github.io/Firmware/>. The clock fetches a small VERSION file from this URL to determine the latest available release; if the server reports a newer version, the corresponding .bin file is downloaded and flashed.

Display Feedback

All update status appears as three-character messages on the matrix. Other display content is suspended for the duration of the update.

Display	Meaning
UPD (yellow)	Connecting to WiFi and the firmware server (3 seconds)
v1.2.3 scrolling green	Software is already current. Returns to normal time display after 3 seconds.
UP1, UP2, ... UP9 cycling	Downloading and flashing new software. The digit cycles every second so the display does not appear frozen during long blocking writes.
OK (green)	Update successful. Reboots automatically after 5 seconds.

Error Codes

If the update fails, the matrix displays a red ERx code, then the clock reboots after 3 seconds:

Code	Meaning
ER1	No WiFi connection or firmware URL not configured
ER2	Could not fetch the VERSION file from the server
ER3	Software download or flash write failed
ER4	VERSION file is malformed (missing or invalid version string)

Version Behavior

Major version change (e.g., 1.x.x to 2.x.x): all settings are reset to factory defaults on the first boot after the update completes. This ensures any new preset slots get sensible defaults instead of leftover bytes from the previous release.

Minor or patch change (e.g., 1.2.3 to 1.2.4): all settings are preserved.

14. Troubleshooting

Symptom	Possible Cause	Solution
"AP" blinks indefinitely	Clock is in AP configuration mode	Connect to the WordClock-5 WiFi and configure at 192.168.4.1.
"WiFi" message persists for more than a few seconds	Wrong WiFi password or out-of-range network	Re-enter AP mode (hold SW2 for 5 s) and reconfigure WiFi.
"SD Card" appears at boot	SD card missing, formatted incorrectly, or no .fmt file in /Language	Insert an SD card formatted FAT32 with the correct format file in the /Language folder.
"WC5" boot message shows wrong colors	Display color order not initialized for this board variant	Power-cycle while holding SW2; release when "WC5" appears. Repeat if still wrong.
Display is dim	Brightness set low, or override period active	Click SW2 to cycle brightness, or press SW1 during the override.
Time is wrong	Incorrect time zone or DST configuration	Check UTC offset and DST start/end settings in AP setup.
Cannot connect to WordClock-5 WiFi	Clock not in AP mode	Hold SW2 for 5+ seconds during normal operation, or apply power with SW1+SW2 held.
LEDs do not light at all	Reverse-voltage protection tripped, or supply voltage too low	Check polarity of the 5 V supply (center positive) and verify the supply is rated 2 A single-board / 4 A multi-board.
LEDs visibly damaged after first power-on	Supply voltage exceeded 5 V	Replace the LEDs and verify the supply rating before reconnecting. The clock is not protected against over-voltage.
Override settings never take effect	Time-zone or DST misconfigured, so the local clock never enters the period's window	Verify UTC offset and DST settings, then re-check override start/end times.
Over-the-air update shows ER1	No WiFi connection or firmware URL not configured	Confirm WiFi is connected and the firmware URL is set in AP configuration.
Over-the-air update shows ER2 / ER3 / ER4	Server unreachable, download failed, or VERSION file malformed	Retry once. If it persists, confirm the firmware URL is correct and the server is reachable from another device on the same network.

Notices

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Operate the clock at the supplied 5 V DC voltage only; higher voltages will damage the LEDs.

WordClock 5 — Designed by Mitchell Feig